MATH 241: Calculus III

Fall 2026

Lecture 15: Applications of Multiple Integrals

Background & Motivation

Now that we can integrate over general regions, we put multiple integrals to work. Mass, center of mass, moments of inertia, and average value all reduce to integrals with the right integrand — the same recipe in 2D and in 3D. We also derive the surface area formula, extending arc length to two dimensions.

1. Warm-Up: Double Integral Review

Evaluate $\displaystyle\int_0^1 \int_0^{1-x} (x+y)\,dy\,dx$.

1. Inner integral (y): $\displaystyle\int_0^{1-x} (x+y)\,dy = \left[xy + \frac{y^2}{2}\right]_0^{1-x} = x(1-x) + \frac{(1-x)^2}{2}$
2. Simplify: $x - x^2 + \frac{1-2x+x^2}{2} = \frac{1}{2} - \frac{x^2}{2}$
3. Outer integral: $\displaystyle\int_0^1 \left(\frac{1}{2} - \frac{x^2}{2}\right) dx = \left[\frac{x}{2} - \frac{x^3}{6}\right]_0^1 = \frac{1}{2} - \frac{1}{6} = \boxed{\frac{1}{3}}$

2. Mass and Center of Mass

Mass, Moments, and Center of Mass

For a lamina with density $\rho(x,y)$ over region R:

$$\text{Mass:}\quad M = \iint_R \rho(x,y)\,dA$$

$$\text{Moments:}\quad M_x = \iint_R y\,\rho(x,y)\,dA \quad \text{(about } x\text{-axis)}$$

$$M_y = \iint_R x\,\rho(x,y)\,dA \quad \text{(about } y\text{-axis)}$$

$$\text{Center of mass:}\quad \bar{x} = \frac{M_y}{M}, \qquad \bar{y} = \frac{M_x}{M}$$

– M_x uses y (distance to x-axis); M_y uses x (distance to y-axis)
– If $\rho = \text{const}$, center of mass = centroid

Center of mass of a system of point masses.

Example 1: Center of Mass of a Triangular Lamina

Find the mass and center of mass of a triangular lamina with vertices $(0,0)$, $(2,0)$, $(0,1)$ and density $\rho(x,y) = 1+x$.

1. **Region (Type I):** $0 \le x \le 2$, $0 \le y \le 1 - x/2$.
2. **Mass:** $M = \int_0^2 \int_0^{1-x/2} (1+x)\,dy\,dx = \int_0^2 (1+x)(1-x/2)\,dx$
$= \int_0^2 \left(1 + \frac{x}{2} - \frac{x^2}{2}\right) dx = \left[x + \frac{x^2}{4} - \frac{x^3}{6}\right]_0^2 = 2 + 1 - \frac{4}{3} = \frac{5}{3}$
3. **Moment M_y:** $M_y = \int_0^2 \int_0^{1-x/2} x(1+x)\,dy\,dx = \int_0^2 x(1+x)(1-x/2)\,dx$
$= \int_0^2 \left(x + \frac{x^2}{2} - \frac{x^3}{2}\right) dx = \left[\frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{8}\right]_0^2 = 2 + \frac{4}{3} - 2 = \frac{4}{3}$
4. **Moment M_x:** $M_x = \int_0^2 \int_0^{1-x/2} y(1+x)\,dy\,dx = \int_0^2 (1+x)\frac{(1-x/2)^2}{2}\,dx = \frac{5}{12}$
5. **Center of mass:** $\bar{x} = \frac{M_y}{M} = \frac{4/3}{5/3} = \frac{4}{5}$, $\qquad \bar{y} = \frac{M_x}{M} = \frac{5/12}{5/3} = \frac{1}{4}$
6. **Result:** $\boxed{(\bar{x}, \bar{y}) = \left(\frac{4}{5}, \frac{1}{4}\right)}$

The center of mass shifts right because the density $1+x$ is heavier on the right.

3. Moments of Inertia

Moments of Inertia

$$I_x = \iint_R y^2\,\rho\,dA \quad \text{(about the } x\text{-axis)}$$
$$I_y = \iint_R x^2\,\rho\,dA \quad \text{(about the } y\text{-axis)}$$
$$I_z = I_0 = \iint_R (x^2+y^2)\,\rho\,dA \quad \text{(polar moment / about the origin)}$$

Note: $I_z = I_x + I_y$.

Physics connections:

- Rotational kinetic energy: $KE = \frac{1}{2}I\omega^2$
- Newton's second law for rotation: $\tau = I\alpha$

Example 2: DJ Turntable Platter

A circular platter (annular disk, inner radius $R_1 = 1$, outer radius $R_2 = 2$) has density $\rho(r) = kr^2$ (denser at the rim). If total mass is $M = 1$ kg, find k and the polar moment of inertia I_z.

1. **Find k from mass:** $M = \int_0^{2\pi} \int_1^2 kr^2 \cdot r\, dr\, d\theta = 2\pi k \int_1^2 r^3\, dr = 2\pi k \left[\frac{r^4}{4}\right]_1^2 = 2\pi k \cdot \frac{15}{4} = \frac{15\pi k}{2}$
$\frac{15\pi k}{2} = 1 \implies k = \frac{2}{15\pi}$
2. **Polar moment of inertia:** $I_z = \iint (x^2 + y^2)\rho\, dA = \iint r^2 \cdot kr^2 \cdot r\, dr\, d\theta$
$I_z = 2\pi k \int_1^2 r^5\, dr = 2\pi k \left[\frac{r^6}{6}\right]_1^2 = 2\pi k \cdot \frac{63}{6} = 21\pi k$
3. **Substitute k:** $I_z = 21\pi \cdot \frac{2}{15\pi} = \frac{42}{15} = \frac{14}{5}$
4. **Result:** $\boxed{I_z = \frac{14}{5} \approx 2.8 \text{ kg} \cdot \text{m}^2}$

Compare: a uniform annular disk of the same mass has $I_z = \frac{1}{2}M(R_1^2 + R_2^2) = 2.5$. *The rim-heavy platter has a larger moment of inertia, as expected.*

4. Surface Area

Theorem 1: Surface Area Formula

The surface area of $z = f(x, y)$ over a region R in the xy-plane:

$$A = \iint_R \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dA$$

– This generalizes the arc length formula $\int \sqrt{1 + (f')^2}\, dx$ to surfaces

Example 3: Surface Area of a Paraboloid

Find the surface area of $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 4$.

1. **Partial derivatives:** $f_x = 2x$, $f_y = 2y$.
2. **Integrand:** $\sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + 4r^2}$ in polar.
3. **Set up (polar):** $A = \int_0^{2\pi} \int_0^2 \sqrt{1 + 4r^2} \cdot r\, dr\, d\theta = 2\pi \int_0^2 r\sqrt{1 + 4r^2}\, dr$
4. **Substitution:** $u = 1 + 4r^2$, $du = 8r\, dr$, so $r\, dr = du/8$.
When $r = 0$: $u = 1$. When $r = 2$: $u = 17$.
$A = 2\pi \cdot \frac{1}{8} \int_1^{17} \sqrt{u}\, du = \frac{\pi}{4} \cdot \left[\frac{2}{3}u^{3/2}\right]_1^{17} = \frac{\pi}{6}(17\sqrt{17} - 1)$
5. **Result:** $\boxed{A = \frac{\pi}{6}(17\sqrt{17} - 1) \approx 36.2}$

5. Applications via Triple Integrals

Mass, Center of Mass, and Moments of Inertia in 3D

For a solid E with density $\rho(x, y, z)$:

$$\text{Mass:} \quad M = \iiint_E \rho\, dV$$

$$\text{Center of mass:} \quad \bar{x} = \frac{1}{M}\iiint_E x\,\rho\, dV, \quad \bar{y} = \frac{1}{M}\iiint_E y\,\rho\, dV, \quad \bar{z} = \frac{1}{M}\iiint_E z\,\rho\, dV$$

$$\text{Moments of inertia:} \quad I_x = \iiint_E (y^2 + z^2)\,\rho\, dV$$

$$I_y = \iiint_E (x^2 + z^2)\,\rho\, dV$$

$$I_z = \iiint_E (x^2 + y^2)\,\rho\, dV$$

– I_x uses $y^2 + z^2$ — the squared distance from the x-axis. Same pattern for I_y and I_z.
– Rotational kinetic energy about the z-axis: $KE = \frac{1}{2} I_z\, \omega^2$.

Example 4: Center of Mass of a Solid Tetrahedron

A solid tetrahedron E has vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ and uniform density $\rho = 1$. Find its center of mass.
1. **Region:** $0 \le z \le 1 - x - y$, $0 \le y \le 1 - x$, $0 \le x \le 1$.
2. **Mass:** $M = \int_0^1\int_0^{1-x}\int_0^{1-x-y} 1\, dz\, dy\, dx = \int_0^1\int_0^{1-x} (1 - x - y)\, dy\, dx = \int_0^1 \frac{(1-x)^2}{2}\, dx = \frac{1}{6}$.
3. **Moment about the yz-plane:** $\iiint_E x\, dV = \int_0^1\int_0^{1-x}\int_0^{1-x-y} x\, dz\, dy\, dx = \int_0^1 x \cdot \frac{(1-x)^2}{2}\, dx = \frac{1}{24}$.
So $\bar{x} = \dfrac{1/24}{1/6} = \frac{1}{4}$.
4. By symmetry of the tetrahedron and density, $\bar{y} = \bar{z} = \frac{1}{4}$.
5. **Result:** $\boxed{(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)}$ — the centroid sits one-quarter of the way along each axis.

Example 5: Moment of Inertia of a Solid Cylinder

A solid cylinder E of radius R and height h sits along the z-axis with uniform density ρ_0. Find I_z in terms of the total mass $M = \pi R^2 h\, \rho_0$.
1. **Cylindrical coordinates:** $0 \le r \le R$, $0 \le \theta \le 2\pi$, $0 \le z \le h$, $dV = r\, dr\, d\theta\, dz$.
2. **Set up:** $I_z = \iiint_E (x^2 + y^2)\rho_0\, dV = \rho_0 \int_0^h\int_0^{2\pi}\int_0^R r^2 \cdot r\, dr\, d\theta\, dz = \rho_0 \cdot h \cdot 2\pi \cdot \frac{R^4}{4}$.
3. **Simplify:** $I_z = \frac{\pi R^4 h\, \rho_0}{2} = \frac{1}{2} M R^2$.
4. **Result:** $\boxed{I_z = \frac{1}{2} M R^2}$ — the standard result for a uniform solid cylinder.

6. Average Value

Average Value of a Function

The average value of f over a region is its integral divided by the size of the region:

$$\text{2D:}\quad \bar{f} = \frac{1}{\text{Area}(R)}\iint_R f\,dA$$
$$\text{3D:}\quad \bar{f} = \frac{1}{\text{Vol}(E)}\iiint_E f\,dV$$

– Same idea as $\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,dx$ from Calc I.

Example 6: Average Temperature in a Cube

The unit cube $E = [0,1]^3$ has temperature $T(x,y,z) = x+y+z$. Find the average temperature.
1. **Volume:** $\text{Vol}(E) = 1$.
2. **Integrate:** $\iiint_E (x+y+z)\,dV = \int_0^1\int_0^1\int_0^1 (x+y+z)\,dz\,dy\,dx$.
By symmetry, $\iiint_E x\,dV = \iiint_E y\,dV = \iiint_E z\,dV = \frac{1}{2}$, so the sum is $\frac{3}{2}$.
3. **Result:** $\bar{T} = \dfrac{3/2}{1} = \boxed{\frac{3}{2}}$ — exactly the value of T at the center $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$, which makes sense because T is linear.

Practice Problems

Problem 1. Find the mass of a lamina over $R = [0,1]\times[0,2]$ with density $\rho(x,y) = xy$.

Answer: $M = 1$

Problem 2. Find I_z for a uniform disk of radius R with constant density ρ_0. Express in terms of the total mass $M = \pi R^2\rho_0$.

Answer: $I_z = \frac{1}{2}MR^2$

Problem 3. A disk of radius 2 has density $\rho = 4 - x^2 - y^2$. A torque of $\tau = 12$ N·m is applied. Find the angular acceleration $\alpha = \tau / I_z$.

Answer: $I_z = 32\pi/3$, so $\alpha = 9/(8\pi) \approx 0.358$ rad/s^2

Problem 4. Find the center of mass of the region bounded by $y = x^2$ and $y = 4$ with uniform density $\rho = 1$.

Answer: $(\bar{x}, \bar{y}) = (0, 12/5)$

Problem 5. Set up (do not evaluate) the surface area integral for $z = e^x \sin y$ over the rectangle $[0,1] \times [0,\pi]$.

Answer: $A = \int_0^1 \int_0^\pi \sqrt{1 + e^{2x}\sin^2 y + e^{2x}\cos^2 y}\, dy\, dx = \int_0^1 \int_0^\pi \sqrt{1 + e^{2x}}\, dy\, dx$

Problem 6. Find the mass of the solid box $E = [0,1] \times [0,2] \times [0,3]$ with density $\rho(x,y,z) = xyz$.

Answer: $M = \int_0^1 \int_0^2 \int_0^3 xyz\, dz\, dy\, dx = \frac{1}{2} \cdot 2 \cdot \frac{9}{2} = \frac{9}{2}$

Problem 7. A solid cone of radius R and height h has uniform density ρ_0 and total mass $M = \frac{1}{3}\pi R^2 h\, \rho_0$. Find its moment of inertia I_z about the central axis. *Hint: cylindrical coordinates with* $0 \le r \le Rz/h$.

Answer: $I_z = \frac{3}{10} M R^2$

Problem 8. Find the average value of $f(x, y, z) = x^2 + y^2 + z^2$ over the unit ball $x^2 + y^2 + z^2 \le 1$. *Hint: spherical coordinates; Vol* $= \frac{4\pi}{3}$.

Answer: $\iiint_E \rho^2 \cdot \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \frac{4\pi}{5}$, so $\bar{f} = \dfrac{4\pi/5}{4\pi/3} = \frac{3}{5}$